

AP Calc Lesson 2.5 Homework

Answer
Key

1. $-4 \sin x + 9$

2. $f(x) = \sin x$ so $f'(x) = \cos x$ and thus $f'\left(\frac{5\pi}{6}\right) = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

3. $a = -1$ and $b = 1 + \frac{\pi}{4}$

4. a. $f'(x) = 1 - 2 \cos x$

b. $y = 3(x - \pi) + \pi$ or equivalent

c. $f'(x) = 0 \Rightarrow 1 - 2 \cos x = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$

5. a. $f'(x) = \cos x + g'(x)$ so $f'(\pi) = \cos \pi + g'(\pi) = -1 - 1 = -2$

b. f' is positive on the interval $\left(0, \frac{\pi}{4}\right)$ since $f'(x) = \cos x + g'(x)$ and $\cos x > 0$ and $g'(x) > 0$ on the interval $\left(0, \frac{\pi}{4}\right)$.

c. $x = \frac{2\pi}{3}$

$f'(x) = 0 \Rightarrow \cos x - \frac{1}{2} = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$, but only $x = \frac{2\pi}{3}$ is in the interval $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$.

6. a. $y = -\frac{1}{2}\left(x - \frac{\pi}{6}\right) + \frac{\sqrt{3}-8}{2}$

$g\left(\frac{\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right) - 4 = \frac{\sqrt{3}}{2} - 4 = \frac{\sqrt{3}-8}{2}$ and $g'(x) = -\sin x$ so $g'\left(\frac{\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}$

b. $x = \frac{2\pi}{3}$

Since the tangent lines are parallel, their slopes must be equal, so $g'(x) = h'(x)$.

Since $h'(x) = \sqrt{3} \cos x$, $-\sin x = \sqrt{3} \cos x \rightarrow \tan x = -\sqrt{3} \Rightarrow x = \frac{2\pi}{3}$ and $x = \frac{5\pi}{3}$. Since x the interval $[0, \pi]$, $x = \frac{2\pi}{3}$.

c. $j(x) = f(x) - 2g(x) = 4x^2 - \sqrt[3]{x} - 2(\cos x - 4) = 4x^2 - \sqrt[3]{x} - 2 \cos x + 8$

$j'(x) = 8x - \frac{1}{3x^{2/3}} + 2 \sin x$

7. a. $h'(x) = \frac{f'(x)}{2} - g'(x) = \frac{f'(x)}{2} - (3 \cos x + 4)$ so $h'(0) = \frac{f'(0)}{2} - 3 \cos(0) - 4 = \frac{-4}{2} - 3(1) - 4 = -2 - 3 - 4 = -9$

b. $\lim_{x \rightarrow \pi^-} j(x) = \lim_{x \rightarrow \pi^-} f(x) = f(\pi) = 3\pi$ and $\lim_{x \rightarrow \pi^+} j(x) = \lim_{x \rightarrow \pi^+} g(x) = \lim_{x \rightarrow \pi^+} (3 \sin x + 4x - \pi) = 3 \sin \pi + 4\pi - \pi = 3\pi$

Since $\lim_{x \rightarrow \pi^-} j(x) = \lim_{x \rightarrow \pi^+} j(x) = j(\pi)$, j is continuous at $x = \pi$.

$\lim_{x \rightarrow \pi^-} j'(x) = \lim_{x \rightarrow \pi^-} f'(x) = 1$ and $\lim_{x \rightarrow \pi^+} j'(x) = \lim_{x \rightarrow \pi^+} g'(x) = \lim_{x \rightarrow \pi^+} (3 \cos x + 4) = 3 \cos \pi + 4 = 3(-1) + 4 = 1$, so j is differentiable at $x = \pi$

8. a. $f'(x) = \frac{1}{\pi} \cos x$
- b. $\frac{1}{\pi} \cos d = \frac{\left(\frac{1}{\pi} \sin d - 2\right) + 3}{d}$
- c. $d = -\pi$
 $f'(d) = \frac{1}{\pi} \cos d$. Since the tangent line passes through the point $(0, -3)$, $f'(d) = \frac{f(d) - (-3)}{d - 0}$. This means
 that $\frac{1}{\pi} \cos d = \frac{\left(\frac{1}{\pi} \sin d - 2\right) + 3}{d} \rightarrow \frac{d}{\pi} \cos d = \frac{1}{\pi} \sin d + 1 \rightarrow d \cos d = \sin d + \pi$. The graph of $y = x \cos x$ intersects the graph of $y = \sin x + \pi$ on the interval $[-4, 4]$ when $x = -\pi$.

9. The slope at Point B is greater because the graph of f is steeper at Point B. The ratio of the slopes

is $\frac{-\sin\left(\frac{3\pi}{2}\right)}{-\sin\left(\frac{7\pi}{6}\right)} = \frac{1}{\frac{1}{2}} = 2$, so the slope at Point B is two times greater than the slope at Point A.

10. A

11. D